

April - July 2026



INTERNSHIP RESULTS

Project Title: Fine-tuning RoBERTa for Classification Tasks

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Laboratory: Scientific Computing Laboratory

“BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”

Authors: Google AI Language, May 2019

Model architecture:

The specifications of two main model sizes are highlighted:

- BERT_BASE: L = 12 layers, H = 768 hidden size, A = 12 attention heads (110 million parameters same as GPT model)
- BERT_LARGE: L = 24 layers, H = 1024, A = 16 attention heads (340 million parameters)

Bidirectionality:

- ELMo - shallowly combines separate left-to-right and right-to-left models) or GPT - strictly left-to-right
- BERT jointly conditions representations on both left and right context in all layers

Masked Language Model (MLM):

To enable bidirectionality the authors introduce MLM. 15% of input tokens are randomly masked, and the model learns to predict the original tokens using only their surrounding context

Next Sentence Prediction (NSP):

The model learns to predict whether sentence B is the actual continuation of sentence A (IsNext) or a randomly sampled sentence

Two-stage training:

- Pre-training: The model learns general language representations on large unlabeled corpora (BooksCorpus and Wikipedia) using MLM and NSP objectives
- Fine-tuning: The pre-trained model is adapted to specific downstream tasks, typically only adding a single output layer

“RoBERTa: A Robustly Optimized BERT Pretraining Approach”

Main problem and motivation

- Impact of hyperparameters - hyperparameter choices have a significant impact on final model performance
- Undertraining - the original BERT model was undertrained

Key modifications

RoBERTa introduces several simple but effective changes to the pretraining procedure compared to BERT:

- Longer training and larger batch sizes
- Removal of NSP objective
- Training on longer sequences: The model is trained using longer text sequences.
- Dynamic masking

New training data (CC-NEWS)

The authors collect a large new dataset called CC-NEWS

Input format and NSP removal.

It is found that removing the NSP loss and using either the FULL-SENTENCES format or DOC-SENTENCES format matches or slightly improves downstream performance. This contradicts the original claims made in the BERT paper.

DATASET DESCRIPTION

Dataset name: Trawling for Trolling

Size: 12,490 samples

Classes:

- Normal (5,053 samples) – No offensive language, trolling, or threats.
- Trolling (4,537 samples) – Content intended to provoke conflict, disrupt discussion, or include mild insults for entertainment purposes.
- Profanity (1,582 samples) – Text containing swear words and vulgar language, but not directed at an individual or group.
- Derogatory (862 samples) – Serious and harmful insults directed at an individual or group, without direct threats or explicit hate speech.
- Hate Speech (456 samples) – Extremely abusive or violent content, including hate speech targeting identity attributes (race, religion, sexual orientation, etc.).

Sources:

Randomly sampled and re-annotated existing popular public datasets: Davidson, Crowdsourcing, Jigsaw, Why Swear, and #MeToo.

Additionally, 1,000 samples were added from the Reddit forum r/RoastMe.

Platforms:

Twitter, Wikipedia Talk Pages and Reddit.

CLASIFICACION ONE - HATE

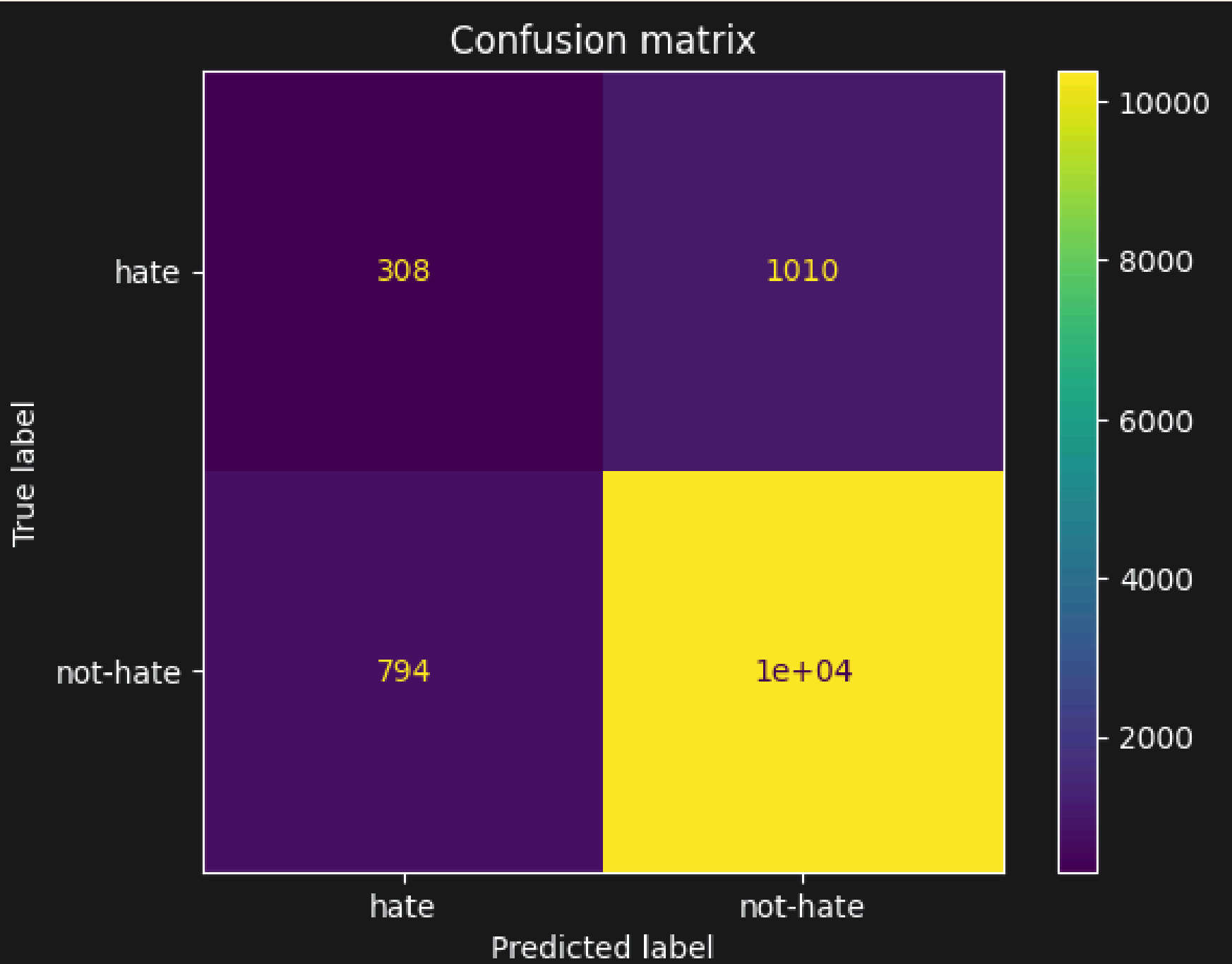
Model name: cardiffnlp/twitter-roberta-base-hate

Model details: roBERTa-base model trained on ~58M tweets and finetuned for hate speech detection

In order to test the model and understand the properties of the dataset we are working with, we make the first experiment.

We do a binary classification of hate and not-hate speech on the entire dataset, where we will group the labels, as written below.

Classification details: not-hate = ["Normal", "Trolling", "Profanity"] and hate = ["Hate","Derogatory"]



```
True negative: 308
False positive: 1010
False negative: 794
True positive: 10378
Accuracy: 0.856
Precision: 0.911
Recall: 0.929
Specificity: 0.234
F1 - score: 0.920
Balanced accuracy: 0.581
```


CLASSIFICATION ONE - offensive

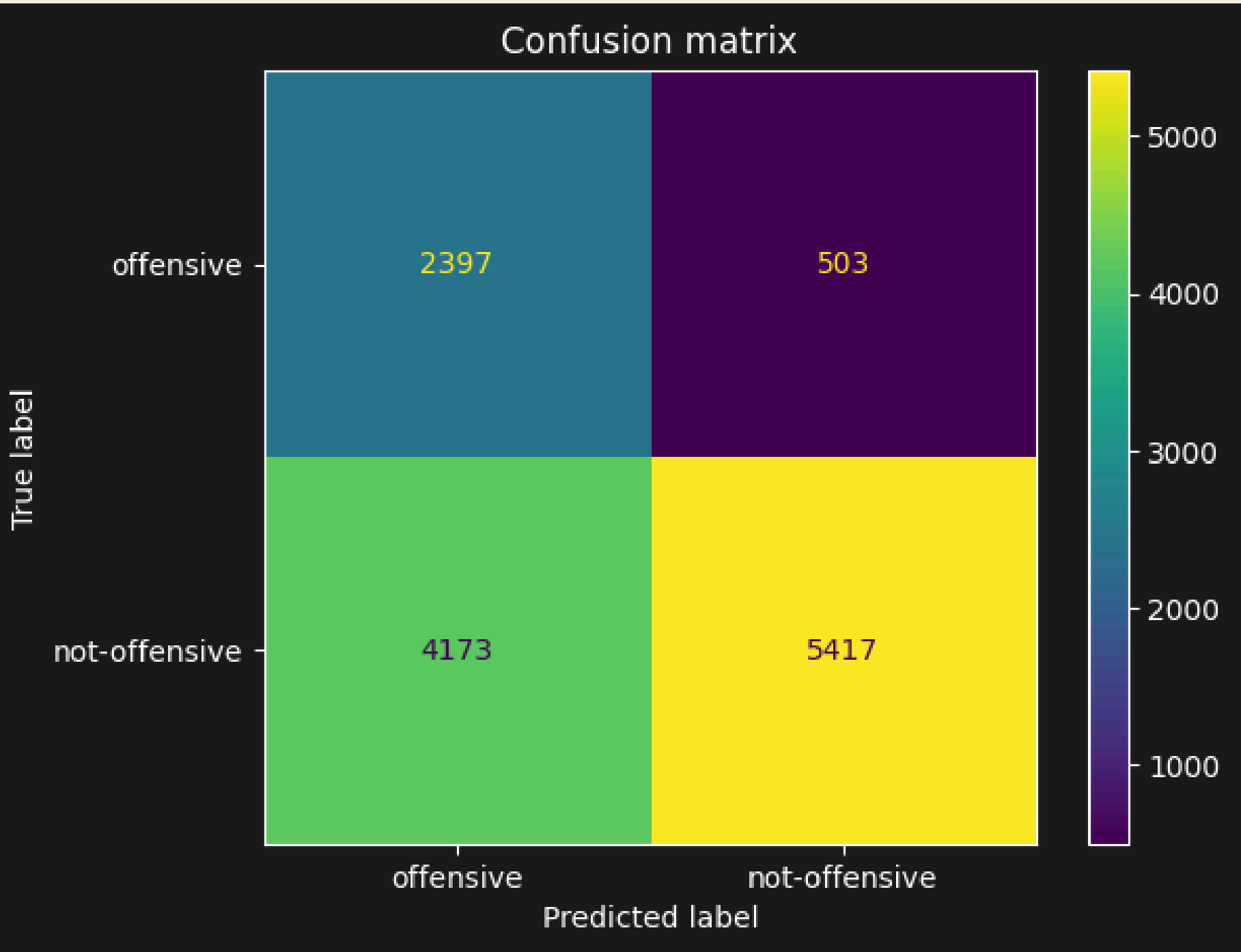
Model name: cardiffnlp/twitter-roberta-base-offensive

Model details: roBERTa-base model trained on ~58M tweets and finetuned for offensive language identification

In order to test the model and understand the properties of the dataset we are working with, we make the first experiment.

We do a binary classification of offensive and not-offensive speech on the entire dataset, where we will group the labels, as written below.

Classification details: not-offensive = ["Normal", "Trolling"] and offensive = ["Hate", "Profanity", "Derogatory"]

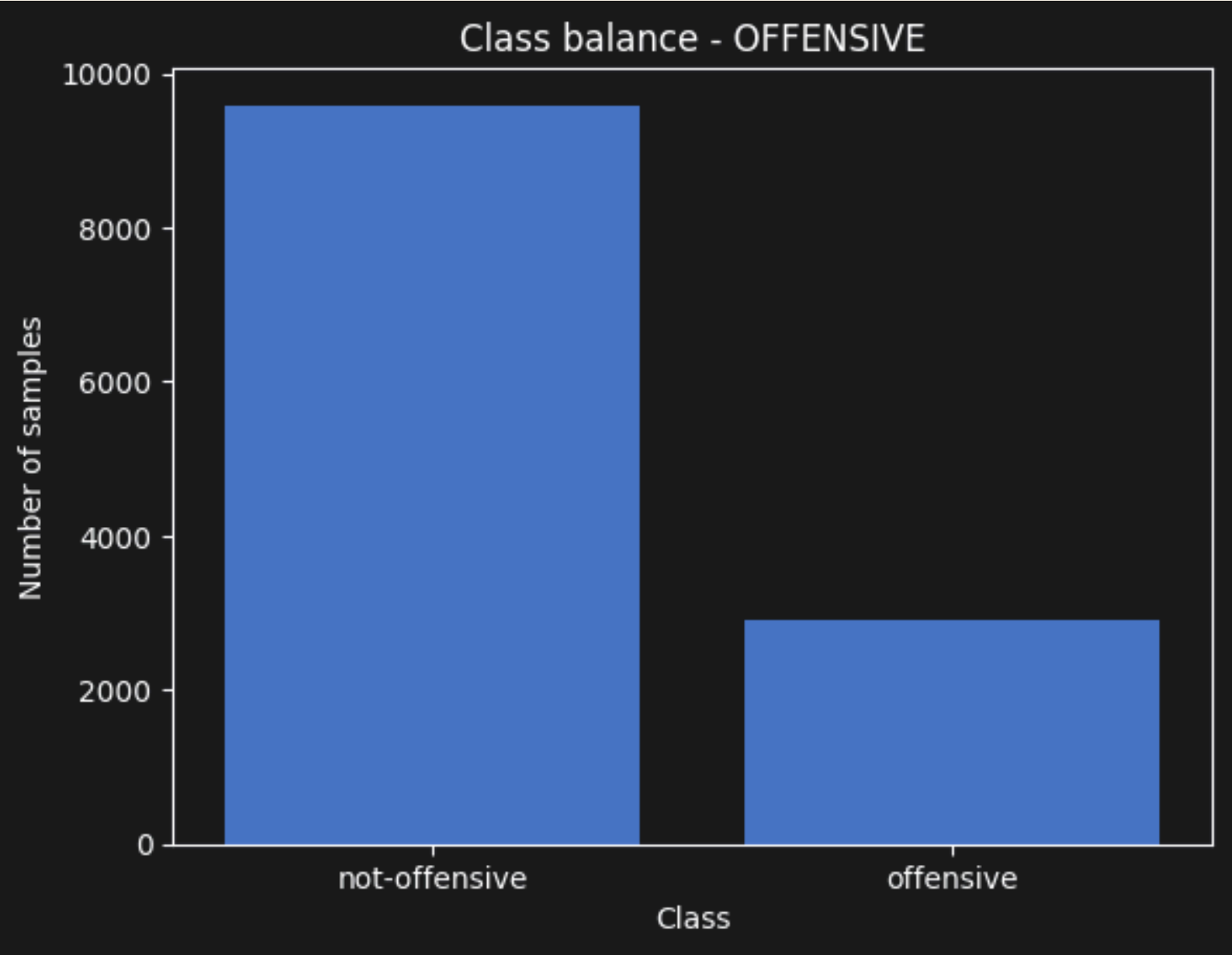


```
True negative: 5417
False positive: 4173
False negative: 503
True positive: 2397
Accuracy: 0.626
Precision: 0.365
Recall: 0.827
Specificity: 0.565
F1 - score: 0.506
Balanced accuracy: 0.696
```

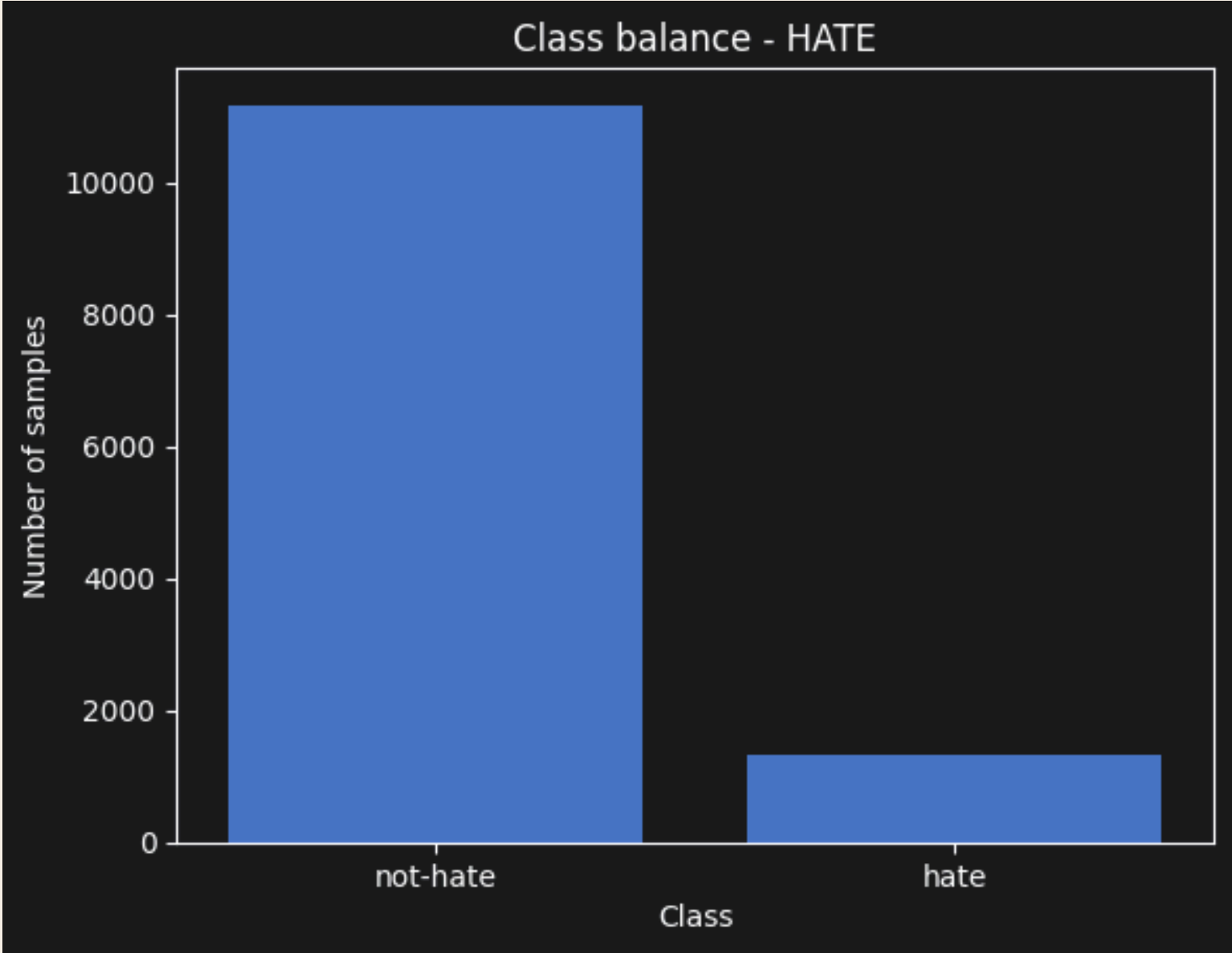
FINE-TUNING AND EVALUATION

BINARY CLASSIFICATION

Fine-Tuning Data preparation

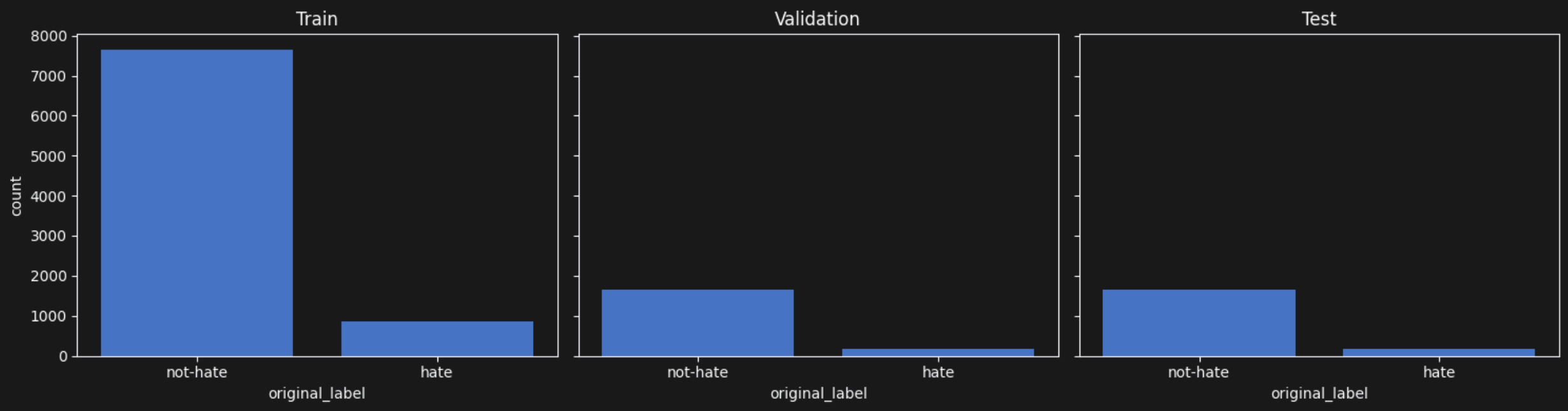


```
OFFENSIVE
Initial number of rows: 12490
After removing exact duplicates: 12252 (Removed: 238)
After removing conflicting samples: 12148 (Conflicting removed: 104)
--- Total retained: 12148 rows (97.26% of original dataset) ---
```



```
HATE
Initial number of rows: 12490
After removing exact duplicates: 12238 (Removed: 252)
After removing conflicting samples: 12162 (Conflicting removed: 76)
--- Total retained: 12162 rows (97.37% of original dataset) ---
```


Fine-Tuning Data split 70\15\15

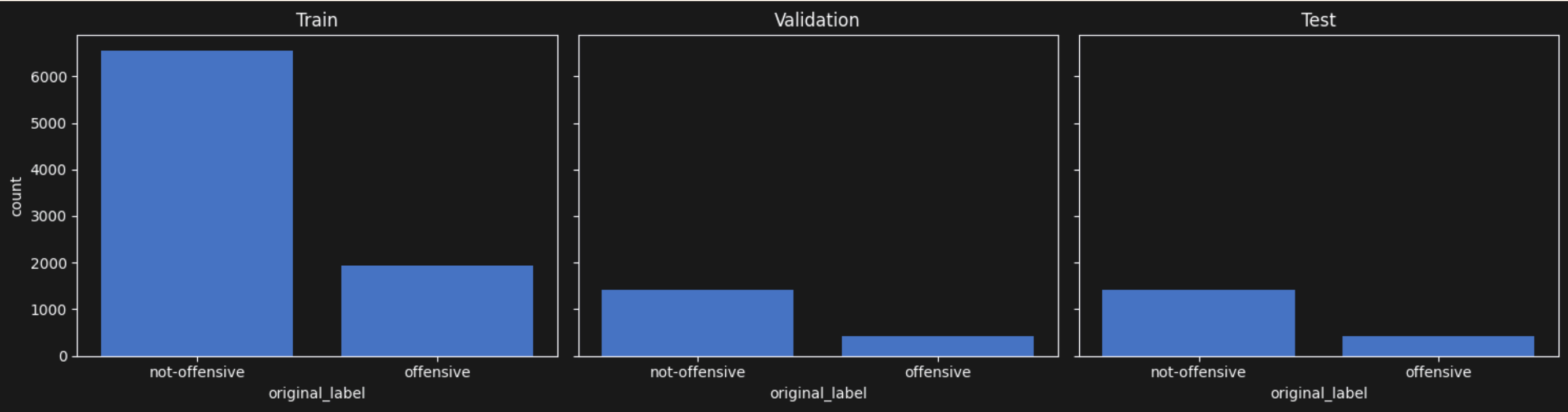


hate

Number of training samples: 8513, not-hate: 7652, hate: 861

Numer of validation samples: 1824, not-hate: 1640, hate: 184

Number of testing samples: 1825, not-hate: 1640, hate: 185



offensive

Number of training samples: 8503, not-offensive: 6556, offensive: 1947

Numer of validation samples: 1822, not-offensive: 1405, offensive: 417

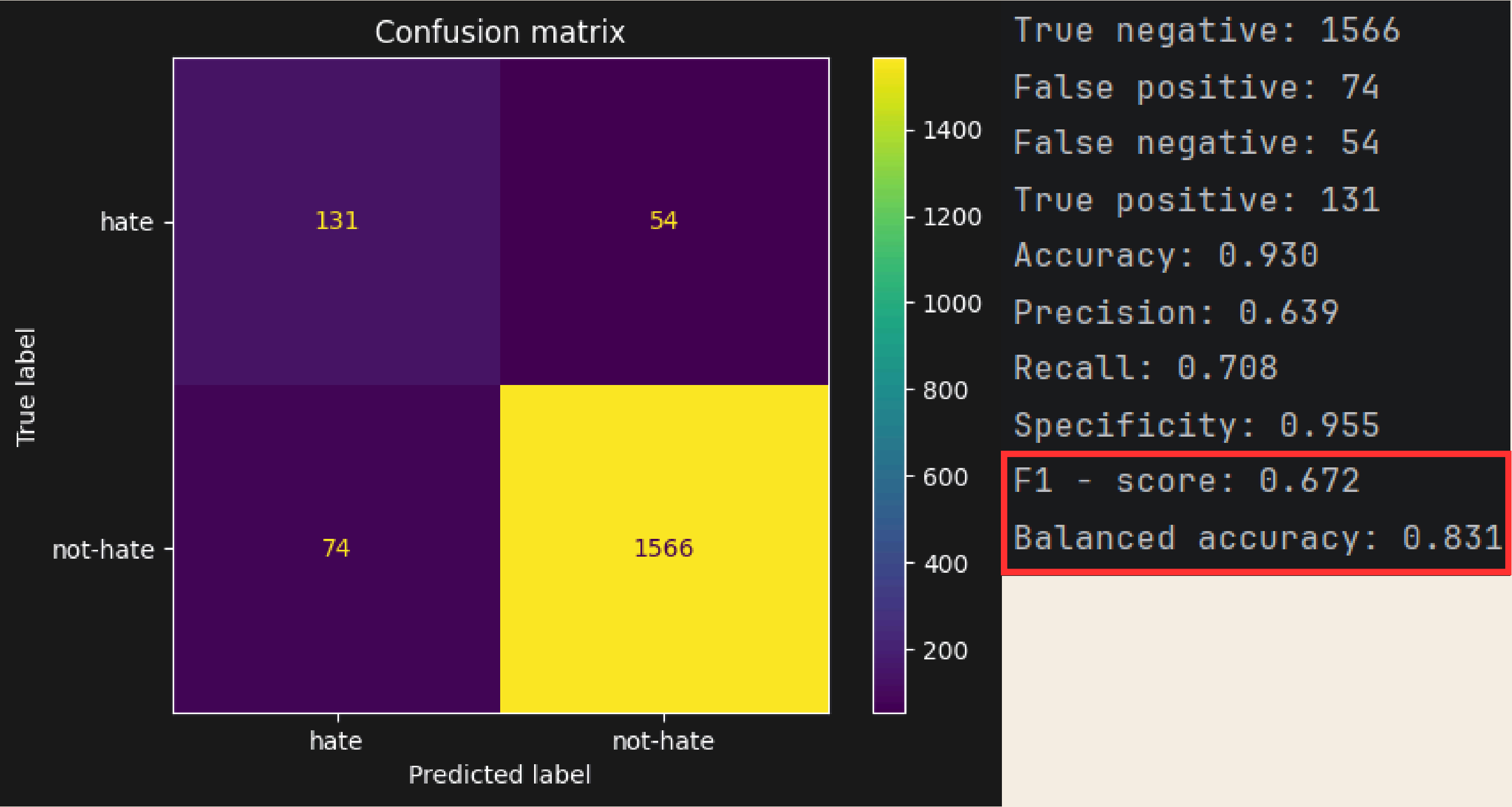
Number of testing samples: 1823, not-offensive: 1406, offensive: 417

Fine-Tunning Hate model

Model name: cardiffnlp/twitter-roberta-base-hate

Loss-function: Weighted Binary Cross Entropy Loss

Class weights: not-hate: 0.556, hate: 4.944



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FALSE POSITIVES (True: 'not-hate', Predicted: 'hate')

First 2: Highest Confidence (close to 1) | Next 2: Borderline Misses (close to 0.5)

****green** and **red** text is for original model, before fine-tuning**

=====

ID: 1358 | Probability: 0.9992

Text: COMPLAIN & AND MORE COMPLAINS THATS WHAT U'RE GOOD FOR !

FUCKIN IDIOT !

If I happen to know who u are I'll wallop the shit out of u damn bastard !

ID: 1358 | True Label: not-hate | Predicted Label: not-hate | Status: Correctly Predicted (True Negative - Not Hate)

ID: 807 | Probability: 0.9988

Text: YOU RACIST!

I'm gonna get the ACLU on you! I am a Latino, and you are reverting my truthful edits because I AM A LATINO! You sick KKK monstor! You and everyone on Wikipedia is gonna sit on a prison if you keep doing this!

ID: 807 | True Label: not-hate | Predicted Label: not-hate | Status: Correctly Predicted (True Negative - Not Hate)

ID: 1159 | Probability: 0.5197

Text: Oh and btw kid's if you have nothing better to do then play wiki wars. I assure you are scum of the earth. We will deal with all of you. Your time is coming.

ID: 1159 | True Label: not-hate | Predicted Label: not-hate | Status: Correctly Predicted (True Negative - Not Hate)

ID: 155 | Probability: 0.5632

Text: Highland BitchK . Them bitches stupid on Bloods , I'd fuck with Deja thoe 👍

ID: 155 | True Label: not-hate | Predicted Label: not-hate | Status: Correctly Predicted (True Negative - Not Hate)

=====

FALSE NEGATIVES (True: 'hate', Predicted: 'not-hate')

First 2: Lowest confidence (close to 0) | Next 2: Borderline misses (close to 0.5)

****green** and **red** text is for original model, before fine-tuning**

=====

ID: 1751 | Probability: 0.0008

Text: Send them all back to the sharia shithole<\b> they came from. #ChristianRefugeesWelcome #MuslimRefugeesNOTWelcome
#IslamSucks

ID: 1751 | True Label: not-hate | Predicted Label: not-hate | Status: Correctly Predicted (True Negative - Not Hate)

ID: 1487 | Probability: 0.0009

Text: Going to sort you out
So you're back eh? I haven't forgotten. You're going to get it you little shit. I have HORRIBLE things planned for you!

ID: 1487 | True Label: not-hate | Predicted Label: not-hate | Status: Correctly Predicted (True Negative - Not Hate)

ID: 268 | Probability: 0.4616

Text: You are very dumb because you offend other users you mother fucker!

ID: 268 | True Label: not-hate | Predicted Label: not-hate | Status: Correctly Predicted (True Negative - Not Hate)

ID: 889 | Probability: 0.4384

Text: Response
YOU ARE LYING AND SLANDERING MY NAME! YOU ARE RACIST AND YOUR BIAS SLURS AND harrassment. I did not vandalize a page you
IDIOT PIECE OF SHIT

ID: 889 | True Label: not-hate | Predicted Label: hate | Status: Incorrectly Predicted (False Positive - Model guessed Hate, but it's Not Hate)

vs BASIC MODEL: cardiffnlp/twitter-roberta-base-hate

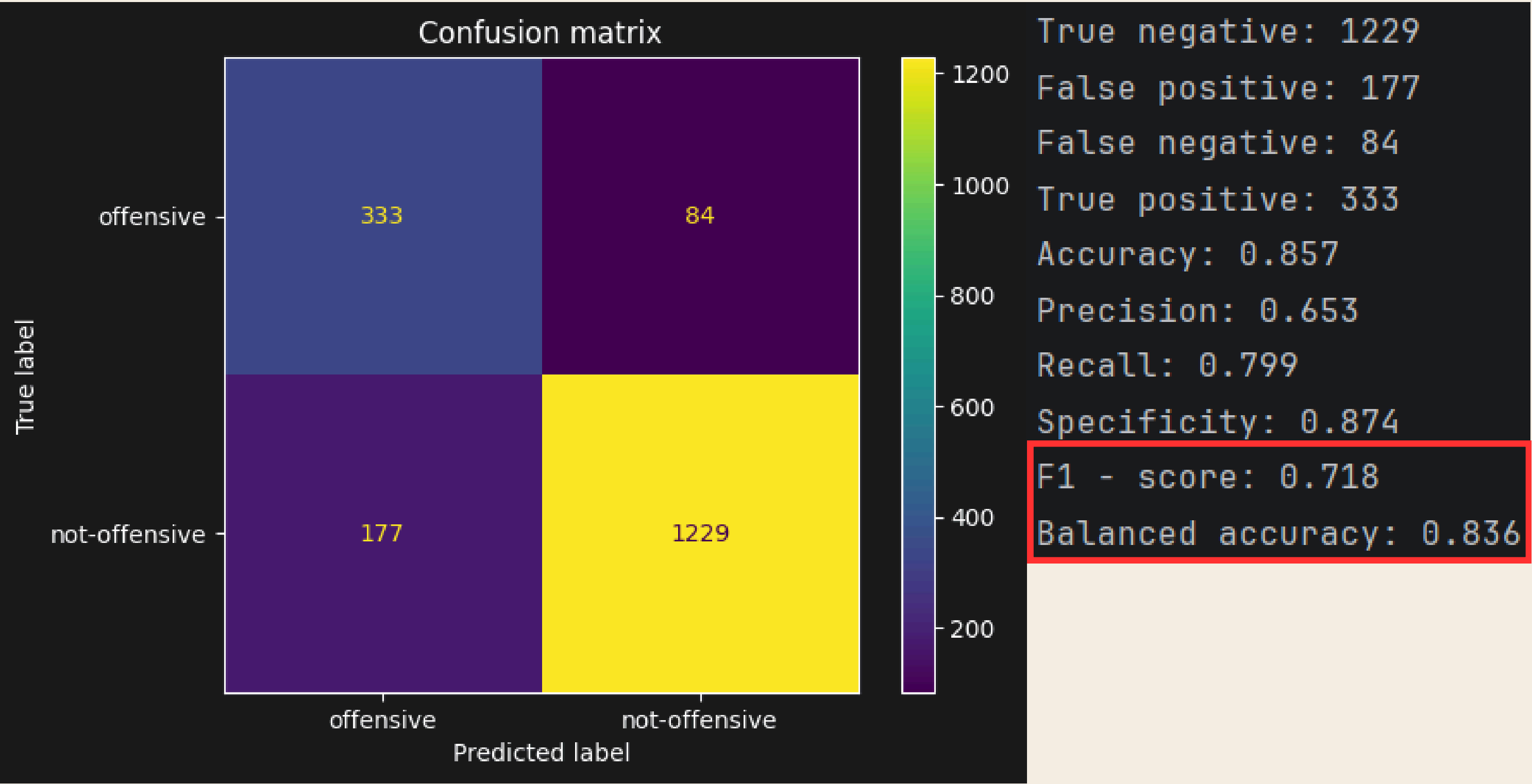
=====			
ID: 155	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 268	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 390	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 515	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 753	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 807	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 889	True Label: not-hate	Predicted Label: hate	Status: Incorrectly Predicted (False Positive - Model guessed Hate, but it's Not Hate)
ID: 1148	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 1159	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 1358	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 1487	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 1562	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 1657	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
ID: 1751	True Label: not-hate	Predicted Label: not-hate	Status: Correctly Predicted (True Negative - Not Hate)
=====			

Fine-Tuning Offensive model

Model name: cardiffnlp/twitter-roberta-base-offensive

Loss-function: Weighted Binary Cross Entropy Loss

Class weights: not-offensive: 0.648, offensive: 2.184



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FALSE POSITIVES (True: 'not-offensive', Predicted: 'offensive')

First 2: Highest Confidence (close to 1) | Next 2: Borderline Misses (close to 0.5)

****green** and **red** text is for original model, before fine-tuning**

=====

ID: 646 | Probability: 0.9950

Text: Bullshit. Go to hell, you fucking vandal.

ID: 646 | True Label: not-offensive | Predicted Label: not-offensive | Status: Correctly Predicted (True Negative - Not Offensive)

ID: 1474 | Probability: 0.9927

Text: <USER> Hey dear I just have to say back in the day MARRIAGE was real and not just a couple of months, stick this fucking marriage

ID: 1474 | True Label: not-offensive | Predicted Label: not-offensive | Status: Correctly Predicted (True Negative - Not Offensive)

ID: 213 | Probability: 0.5034

Text: FUCKKKKKKKKKKKK FUCKKKKKKKKKKKK USA FUCKKKKKKKKKKKKKKKKK

ID: 213 | True Label: not-offensive | Predicted Label: not-offensive | Status: Correctly Predicted (True Negative - Not Offensive)

ID: 1363 | Probability: 0.5094

Text: "
One fuck a day. Got it. Now to convince the wife... X "

ID: 1363 | True Label: not-offensive | Predicted Label: offensive | Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)

=====

FALSE NEGATIVES (True: 'offensive', Predicted: 'not-offensive')

First 4: Lowest Confidence (close to 0) | Next 4: Borderline Misses (close to 0.5)

****green** and **red** text is for original model, before fine-tuning**

=====

ID: 945 | Probability: 0.0017

Text: Silly guy. You should use email like the wikijews.

ID: 945 | True Label: not-offensive | Predicted Label: offensive | Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)

ID: 1204 | Probability: 0.0076

Text: You're fucking crazy!!! The DCEU needed no saving. <https://t.co/mXOeDjLA1s>

ID: 1204 | True Label: not-offensive | Predicted Label: offensive | Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)

ID: 1201 | Probability: 0.4717

Text: FUCK OFF YOUR PIECE OF SHIT JEW

ID: 1201 | True Label: not-offensive | Predicted Label: offensive | Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)

ID: 744 | Probability: 0.4600

Text: Females stay mad at me and I be like por que... don't sweat me cause I'm sure as hell ain't sweating you.

ID: 744 | True Label: not-offensive | Predicted Label: not-offensive | Status: Correctly Predicted (True Negative - Not Offensive)

vs BASIC MODEL: cardiffnlp/twitter-roberta-base-offensive

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OFFENSIVE CLASS PREDICTION EVALUATION FOR TARGET IDs

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ID: 213	True Label: not-offensive	Predicted Label: not-offensive	Status: Correctly Predicted (True Negative - Not Offensive)
ID: 541	True Label: not-offensive	Predicted Label: not-offensive	Status: Correctly Predicted (True Negative - Not Offensive)
ID: 646	True Label: not-offensive	Predicted Label: not-offensive	Status: Correctly Predicted (True Negative - Not Offensive)
ID: 744	True Label: not-offensive	Predicted Label: not-offensive	Status: Correctly Predicted (True Negative - Not Offensive)
ID: 945	True Label: not-offensive	Predicted Label: offensive	Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)
ID: 1071	True Label: offensive	Predicted Label: offensive	Status: Correctly Predicted (True Positive - Offensive)
ID: 1201	True Label: not-offensive	Predicted Label: offensive	Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)
ID: 1203	True Label: not-offensive	Predicted Label: offensive	Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)
ID: 1204	True Label: not-offensive	Predicted Label: offensive	Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)
ID: 1363	True Label: not-offensive	Predicted Label: offensive	Status: Incorrectly Predicted (False Positive - Model guessed Offensive, but it's Not Offensive)
ID: 1451	True Label: not-offensive	Predicted Label: not-offensive	Status: Correctly Predicted (True Negative - Not Offensive)
ID: 1474	True Label: not-offensive	Predicted Label: not-offensive	Status: Correctly Predicted (True Negative - Not Offensive)
ID: 1476	True Label: not-offensive	Predicted Label: not-offensive	Status: Correctly Predicted (True Negative - Not Offensive)
ID: 1800	True Label: not-offensive	Predicted Label: not-offensive	Status: Correctly Predicted (True Negative - Not Offensive)

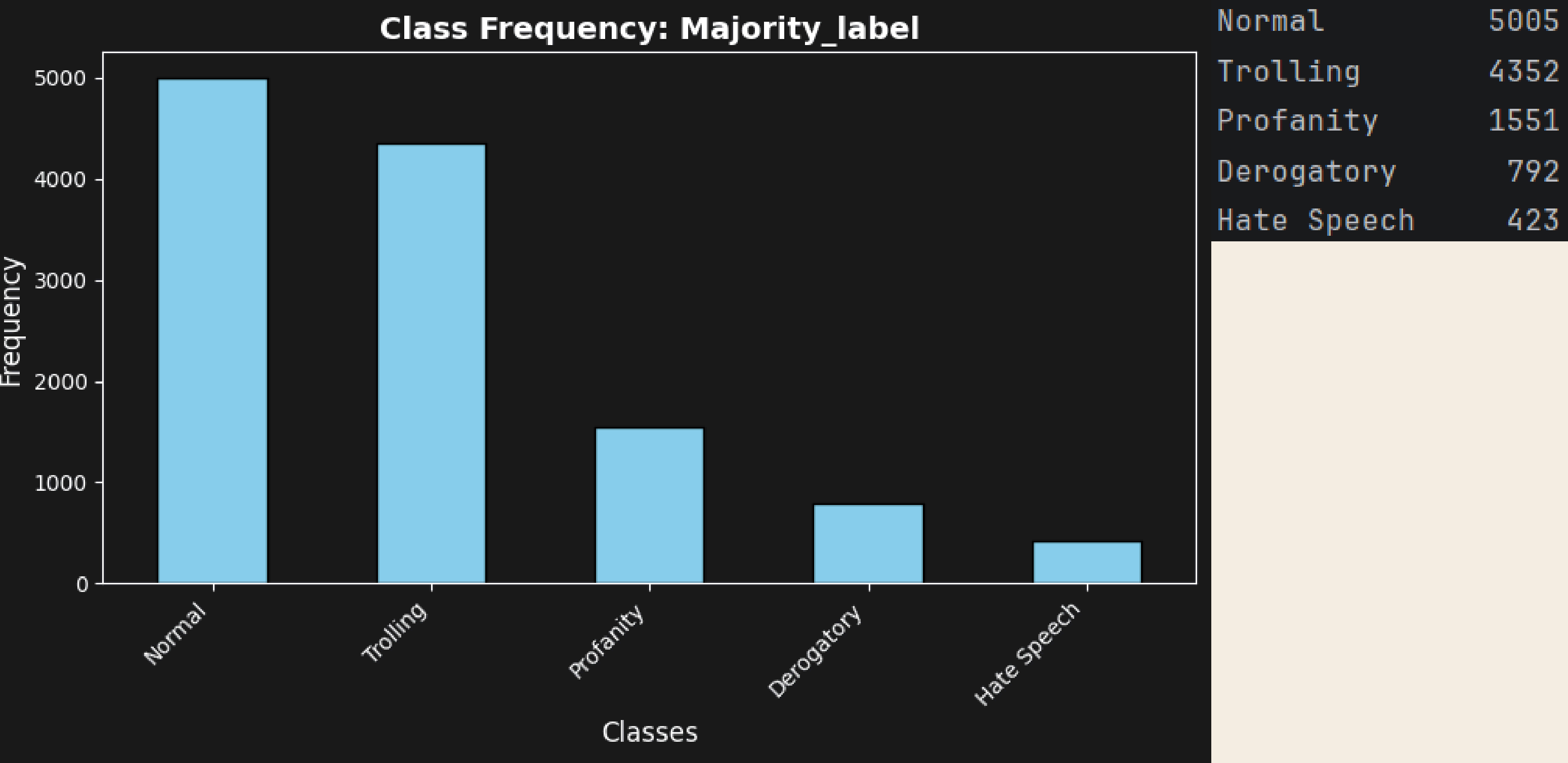
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FINE-TUNING AND EVALUATION

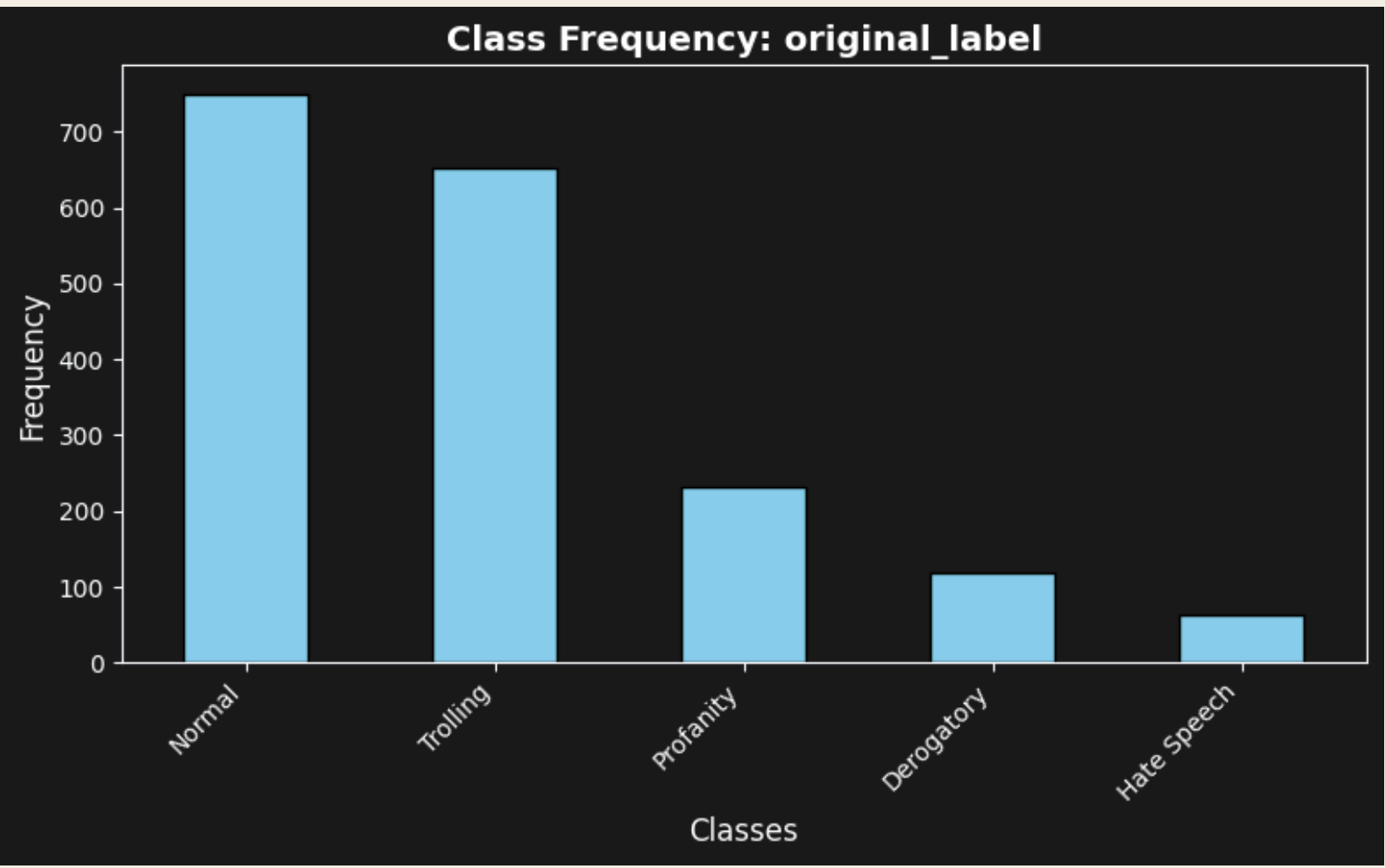
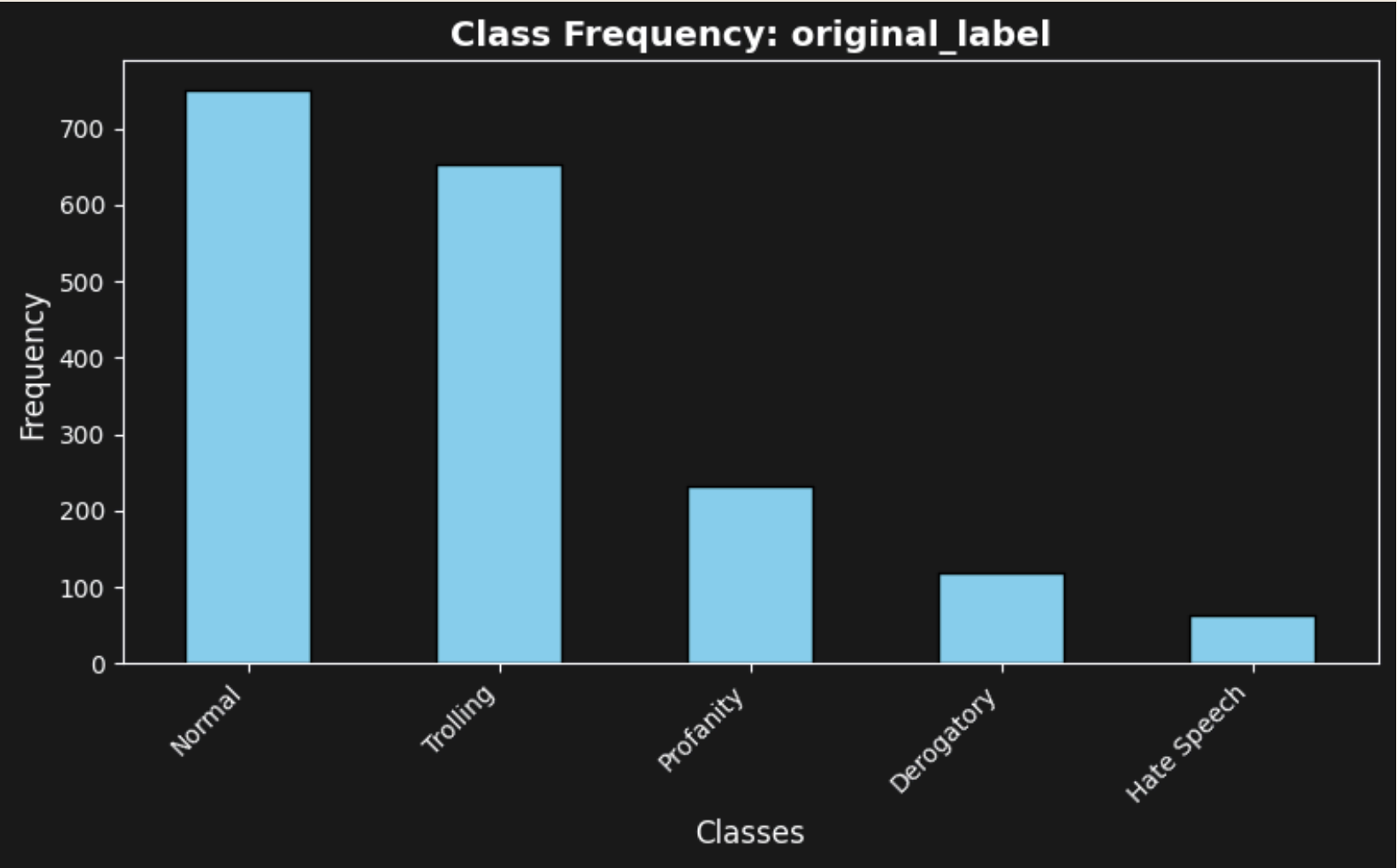
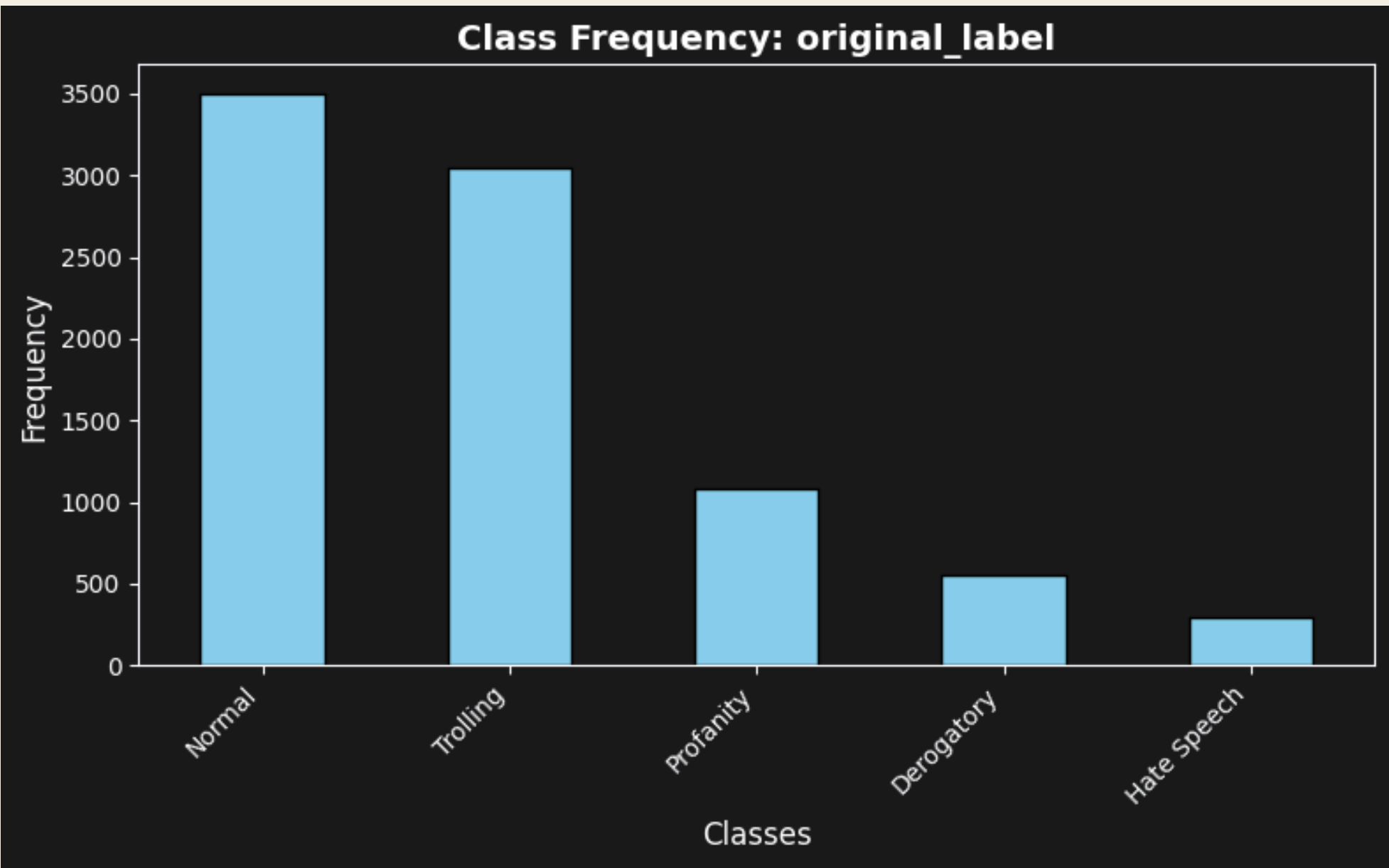
FIVE CLASS CLASSIFICATION

Fine-Tunning Data preparation

```
Initial number of rows: 12490
After removing exact duplicates: 12277 (Removed: 213)
After removing conflicting samples: 12123 (Conflicting removed: 154)
--- Total retained: 12123 rows (97.06% of original dataset) ---
```



Fine-Tuning Data split 70\15\15

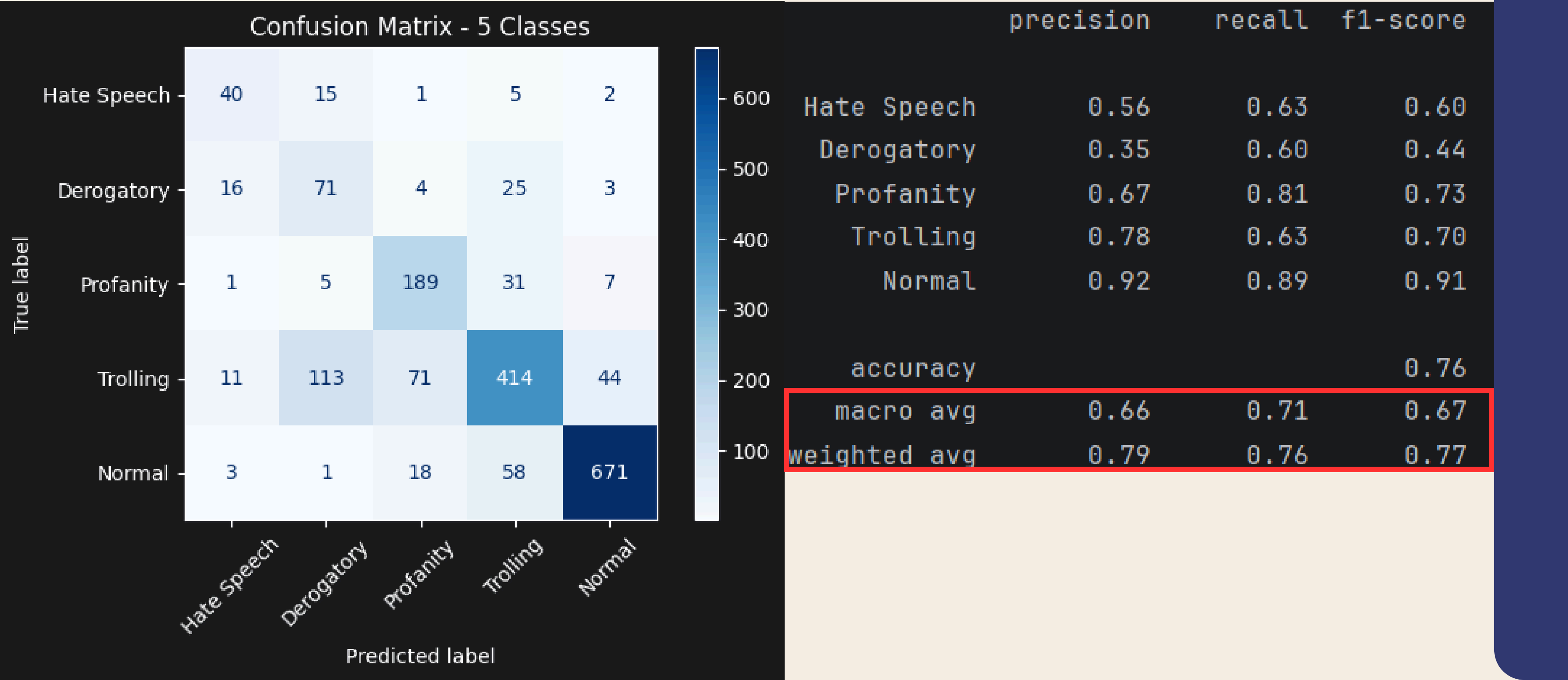


Fine-Tunning Five class model

Model name: roberta-base

Loss-function: Weighted Cross Entropy Loss

Class weights: Hate: 5.734, Derogatory: 3.065, Profanity: 1.563, Trolling: 0.557, Normal: 0.484



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ERROR ANALYSIS (5 CLASSES) - Examples with Highest Predicted Probability

=====

PAIR: [True: Trolling] / [Predicted: Derogatory]

ID: 1559 | Predicted Probability: 0.9552

Text: because I am a huge faggot

.....

ID: 1137 | Predicted Probability: 0.9487

Text: Screw you motherfucker!!! 80.220.6.195

.....

PAIR: [True: Trolling] / [Predicted: Profanity]

ID: 1267 | Predicted Probability: 0.9768

Text: RT @extaeordinary: NIOBE HOW CAN YOU FUCKING SURVIVE THIS <https://t.co/4vbf23KWgX>

.....

ID: 700 | Predicted Probability: 0.9766

Text: RT @YaaBoyShiki: How the fuck can some people not get atleast 1 on flappy bird. I just got 11 while smoking a fuckin square.

.....

PAIR: [True: Trolling] / [Predicted: Normal]

ID: 583 | Predicted Probability: 0.9883

Text: Bigots demand recount in #Idaho #gay rights ordinance referendum <http://t.co/TorYEXxMwi> #tcot #teaparty #teabaggers #lgbt

.....

ID: 144 | Predicted Probability: 0.9883

Text: Black magic😂😂😂 kuchh to believable story bana leti!! #MeToo #FeminismIsCancer #MenToo <https://t.co/wulKVyycfw>

.....

PAIR: [True: Normal] / [Predicted: Trolling]

ID: 1300 | Predicted Probability: 0.9431

Text: Damn dude.

You're in the wrong sub.

This is Roast me, not Roost me.

.....

ID: 545 | Predicted Probability: 0.9407

Text: you look like someone from kpop

.....

PAIR: [True: Profanity] / [Predicted: Trolling]

ID: 896 | Predicted Probability: 0.9099

Text: "

Now I get your problem. You really don't understand what you're being told, so when you call someone an ""ignorant ass"" you think that your own self was just mocked.

First, did I call it confusing? No.

I said that you just mentioned it for some reason with no relation to the way it was told or the situation, and I said that it was a reply to you, saying that vandalizing, as you accused me when reverting my contributions, by writing ""Cum, Leave"" makes no sense.

And your ""WP status"" is relevant as I want to know who the hell are you, and what do you want and think when you keep submitting blocks to my user, as well for keeping reverting MY talk page when I'm trying to get it clear. I don't need this ""discussion"" and I wasn't talking to you.

And I'm not an ""editor"", all I did was improving that article I've seen, and that's why you're even more annoying and less helpful. "

.....

ID: 1227 | Predicted Probability: 0.9079

Text: @Mew2th @MewtwoDiscord @MistakeZss insulting mistake for being "carried" and ranting on discord for forever. You dont know shit son

CONCLUSION

- **Offensive Model Progress:** Fine-tuning directly and significantly upgraded the offensive speech model, boosting both its F1-score (from 0.506 to 0.718) and Balanced Accuracy (from 0.696 to 0.836).
 - **Offensive Model Behavior:** This double increase confirms the offensive classifier moved from an uncertain, high-error state to a highly precise and balanced predictor.
 - **Hate Model Baseline:** Before fine-tuning, the hate speech model was highly biased toward the majority class, displaying a deceptively high F1-score of 0.92 but a near-random Balanced Accuracy of just 0.58.
 - **Hate Model Correction:** Post-fine-tuning, this bias was successfully corrected as the model traded its artificial F1-score for a realistic 0.67 while its Balanced Accuracy surged to 0.831, proving it finally learned to genuinely identify the rare minority class.
-
- **Deceptive Overall Accuracy:** Exceptional performance on the "Normal" class (F1: 0.91) masks weaker areas, inflating the overall accuracy to a deceptive 0.76.
 - **Confusion Among Toxic Classes:** The model struggles with nuances of toxicity, particularly with "Derogatory" (lowest F1: 0.44), heavily misclassifying 113 "Trolling" comments as "Derogatory."